pRisk-Pressure: A Pressure-Conditioned Probabilistic Metric for Evaluating LLM Safety

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Abstract

Current safety evaluations of large language models (LLMs) commonly report jailbreak success rates as a function of adversarial query budget or attack iterations. While these measurements provide empirical evidence of vulnerability under increasing optimization effort, they remain largely descriptive: they report observed attack outcomes rather than modeling the underlying likelihood of failure as adversarial pressure scales.

We introduce **pRisk-Pressure**, a framework that treats unsafe behavior as a stochastic response to bounded adversarial optimization. By estimating risk-pressure curves, we characterize how the probability of failure evolves across various attack strategies and pressure levels. Using standardized benchmarks across model scales, we demonstrate that conventional success rates fail to capture real-world dynamics where attackers persist until success. pRisk-Pressure shifts evaluation from empirical demonstration to predictive, likelihood-based measurement, offering a more faithful characterization of model robustness. Evaluating models across parameter scales, reasoning capabilities, and training stages, we show that conventional pointwise success rates obscure systematic differences in robustness and fail to reflect real-world adversarial dynamics, where attackers adapt and persist until success. In contrast, pRisk-Pressure estimates the latent likelihood of failure under sustained optimization, providing a principled and more faithful characterization of real-world exploitability.

1 Introduction

Safety benchmarks for LLMs commonly report aggregate jailbreak success rates under fixed adversarial prompts [Carlini et al., 2023, Wei et al., 2023]. This binary reporting paradigm answers whether a model *can* fail, but not how failure likelihood evolves under increasing adversarial effort.

In practice, adversaries iteratively refine prompts, adapt strategies, and apply search procedures. A model that appears robust under naive prompting may become vulnerable under moderate optimization pressure.

We argue that safety should not be evaluated at a single operating point. Instead, it should be modeled as a probabilistic response function under adversarial scaling.

To this end, we introduce **pRisk-Pressure**, a framework that estimates:

$$R(\lambda) = \Pr(\text{unsafe} \mid \lambda),$$

where λ denotes adversarial optimization pressure.

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This transforms safety evaluation from static demonstration to pressure-conditioned likelihood estimation.

2 Related Work

Evaluating the safety of large language models has become an active area of research as models are increasingly deployed in high-stakes and open-ended contexts. A broad class of work focuses on jailbreaking and adversarial prompting, where crafted inputs are designed to circumvent safety alignments and elicit harmful or policy-violating outputs. Benchmarks such as JailbreakBench [Chao et al., 2024] provide standardized datasets of adversarial prompts and evaluation frameworks to systematically compare jailbreak success rates and defenses across models, addressing issues of reproducibility and varied scoring methods that earlier ad-hoc approaches lacked. Research on dynamic jailbreak attacks, for example, h4rm3l [Doumbouya et al., 2025], which composes parameterized attack primitives to automatically generate novel jailbreaks with high success rates, highlights the limitations of static prompt sets and the need for composable evaluation strategies. Beyond outright jailbreaks, studies have examined how ethical biases in models can be exploited to induce unsafe outputs, indicating broader vulnerabilities tied to model training and alignment objectives [Lee and Seong, 2025]. Surveys and adversarial robustness frameworks also underscore the challenge of assessing safety across languages and transformation types, and the lack of consensus on risk metrics beyond binary failure rates [Cantini et al., 2025].

In parallel, the rapid evolution of agentic AI, where LLMs serve as autonomous agents that interact with tools and environments over multiple steps, has motivated dedicated benchmarks and analyses of emergent risk in dynamic settings. AgentHarm [Andriushchenko et al., 2025] is a recently proposed benchmark that measures the harmfulness of LLM agents executing multi-stage tasks across diverse categories such as fraud and cybercrime, revealing that models often comply with malicious requests and retain capabilities post-jailbreak. Other benchmarks like R-Judge [Yuan et al., 2024] evaluate risk awareness in multi-turn agent interaction records, while Agent-SafetyBench [Zhang et al., 2025] assesses broad safety failure modes across hundreds of interactive environments and test cases. More general surveys of agent evaluation methods identify gaps in current benchmarks for planning, tool use, and robustness, signaling a need for richer, more adaptive evaluation frameworks that reflect the complexity of real-world dynamic tasks [Yehudai et al., 2025]. Collectively, these works motivate a shift from binary, single-shot safety evaluation toward probabilistic, likelihood-based measures that quantify risk across perturbations and interaction patterns.

3 Benchmark Selection

We evaluate our framework using two complementary safety benchmarks:

JailbreakBench: Focused on adversarial jailbreak prompts and structured attack templates [Chao et al., 2024].

HarmBench: A broader harm taxonomy covering diverse unsafe behaviors beyond jailbreak framing [Mazeika et al., 2024].

Using two benchmarks allows us to test both optimization-heavy and taxonomy-driven safety settings while maintaining experimental depth. Due to resource constraints, we use a subset of up to 100 prompts from each source.

4 Operationalizing Optimization Pressure

We define optimization pressure λ as a bounded adversarial refinement budget under a fixed attack procedure.

Let λ denote the maximum number of refinement iterations allowed.

We evaluate discrete levels:

$$\lambda \in \{0, 1, 3, 5, 10, ...\text{n}\}.$$

Algorithm 1: Budgeted Iterative Refinement for Prompt Optimization

Input: Base prompt p; target model M; unsafe evaluator $E(\cdot) \in \{0, 1\}$; refinement policy π; budget λ
Output: Final prompt $p^{(\lambda)}$; success flag s; first-success step $t^\star$

$p^{(0)} \leftarrow p$; $s \leftarrow 0$; $t^\star \leftarrow \perp$;
for $t = 1$ **to** λ **do**
  $y^{(t-1)} \leftarrow M\big(p^{(t-1)}\big)$;
  $z^{(t-1)} \leftarrow E\big(y^{(t-1)}\big)$;
  if $z^{(t-1)} = 1$ ***and*** $s = 0$ **then**
    $s \leftarrow 1$; $t^\star \leftarrow t - 1$;
  $p^{(t)} \leftarrow \pi\big(p^{(t-1)}, y^{(t-1)}, z^{(t-1)}, t\big)$;
return $\big(p^{(\lambda)}, s, t^\star\big)$;

- $\lambda = 0$: Base benchmark prompt (no optimization).
- $\lambda > 0$: Iterative black-box refinement with bounded query budget.

The attack algorithm is fixed across all models and pressure levels. Only the allowed refinement budget varies, ensuring comparability.

5 Risk Definition

For model M, prompt p, and pressure level λ:

$$\hat{R}(M, \lambda) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\text{unsafe}(M(\phi_\lambda(p_i)))].$$

The sequence $\{\hat{R}(M, \lambda)\}$ forms the model's **risk-pressure curve**. Here, $\phi_\lambda(p_i)$ denotes the refined prompt after λ optimization steps under the fixed refinement policy π. We compute bootstrap confidence intervals to quantify statistical uncertainty.

We operationalize risk using a strong pretrained LLM as an automated judge: for each candidate output, we prompt the judge with a clear safety rubric that defines harmful versus compliant behavior, and the judge assigns a binary or graded safety label accordingly.

6 Iterative Refinement Algorithm

We operationalize optimization pressure via a fixed, black-box iterative refinement procedure with a bounded budget λ, which is number of queries. The procedure proposes candidate prompt variants and uses an unsafe evaluator to guide refinement. Importantly, the refinement policy is held constant across models; only the budget varies.

6.1 Refinement Policies and Implementation

All refinement policies π used in this study are instantiated within the *EasyJailbreak* [Zhou et al., 2024] framework, which provides standardized, reproducible implementations of several state-of-the-art jailbreak algorithms under a unified interface. Using a single framework ensures consistent prompt formatting, evaluation logic, and budget control across all experiments.

We focus on a small set of representative refinement policies implemented in EasyJailbreak [Zhou et al., 2024]:

We select **JailBroken** [Wei et al., 2023], **GCG** [Zou et al., 2023], and **PAIR** [Chao et al., 2025] to represent three complementary jailbreak paradigms. JailBroken captures surface-level obfuscation attacks, GCG models optimization-driven adversarial search, and PAIR exploits semantic and reasoning-level manipulation. Together, they provide diverse coverage of syntactic, algorithmic, and cognitive attack strategies for evaluating model robustness. Rather than exhaustively comparing all available jailbreak methods, our goal is to use a small set of representative, state-of-the-art refinement strategies to characterize how unsafe behavior scales under bounded adversarial effort.

7 Summary Metrics

From the risk-pressure curve, we compute:

7.1 Area Under Risk Curve (AURC)

$$\text{AURC}(M) = \sum_{\lambda} R(M, \lambda)\Delta\lambda$$

Captures overall exploitability.

7.2 Optimization Sensitivity

$$\Delta R = R(M, \lambda_{\max}) - R(M, 0).$$

Measures risk increase under adversarial scaling.

7.3 Pressure Threshold

$$\lambda^* = \min\{\lambda : R(M, \lambda) \geq \tau\},$$

where τ is a predefined safety tolerance.

8 Statistical Analysis

For each attack plan, we analyze interaction effects using mixed-effects regression:

$$\text{Risk} \sim \text{Pressure} \times \text{Model}$$

We test for statistically significant differences in:

- Risk curve slope,
- AURC,
- ΔR across models.

9 Models

We evaluate widely used, state-of-the-art models spanning open-source and closed-source systems with varying parameter sizes from different training stages:

- **Qwen-2.5 Family (Alibaba, Open-source)** — multiple scales with base and instruct variants:
  - Qwen-2.5-0.5B-Instruct
  - Qwen-2.5-3B-Instruct
  - Qwen-2.5-7B-Instruct
- **Llama-3 Family (Meta, open-source)** — multiple scales with base vs instruct:
  - Llama-3.2-1B (Base, and Instruct)
  - Llama-3.1-8B (Base, and Instruct)
- **Tulu-2 Family (Ai2, Open-source)** — isolates training/stage effects:

  - Tulu-2-7B (Base, SFT, DPO, and RLVR)
- **Closed-Source (Proprietary) Models** — representing advanced safety-tuned systems:
  - GPT-4o-mini (OpenAI)
  - Claude-3.5-Sonnet (Anthropic)
  - Gemini Flash Lite (Google DeepMind)

Note: The model list may be revised based on resource availability and computational constraints.

10 Ablation Studies

To validate the functionality of **pRisk-Pressure**, we conduct a structured set of ablation studies across pressure level, attack strategy, model scale, training stage, and deployment regime (open vs. closed). Each ablation isolates one axis while holding others fixed, enabling controlled analysis of how different factors shape pressure-conditioned risk likelihood.

Note: We outline potential ablation studies ranked by priority; however, depending on time and resource constraints, we may only complete a subset of them.

10.1 Pressure Sensitivity

We analyze how risk evolves as adversarial pressure increases. For discrete pressure levels $\lambda \in \{0, 1, 3, 5, 10, \dots\}$, we estimate full risk-pressure curves and compute summary metrics including Area Under the Risk Curve (AURC), optimization sensitivity ΔR, and pressure threshold λ^*.

Q: Does risk increase monotonically with pressure across models?

Q: Does single-point reporting at $\lambda = 0$ systematically underestimate exploitability?

10.2 Attack Plan Sensitivity

We evaluate multiple fixed attack strategies (e.g., JailBroken, GCG, PAIR) while holding the pressure schedule constant.

Q: Are risk curves stable across attack plans, or do models exhibit plan-specific vulnerabilities?

10.3 Scale Effects Within Families

[Optional (if time permits)]

Using multiple sizes within the same architecture (e.g., Qwen-2.5 0.5B/3B/7B), we isolate parameter scale while controlling for model family.

Q: Does increasing model size reduce overall risk, or primarily shift the pressure threshold?

10.4 Training Stage Effects

[Optional (if time permits)]

We compare base, instruction-tuned (SFT), and preference-optimized variants (e.g., DPO, RLVR) within shared backbones such as Tulu-2.

Q: Do later alignment stages meaningfully reduce optimization sensitivity?

10.5 Open vs. Closed Systems

[Optional (if time permits)]

Finally, we compare open-source models with proprietary systems (e.g., GPT-4o-mini, Gemini Flash Lite, Claude-3.5-Sonnet).

Q: Do closed-source models exhibit structurally different risk-pressure profiles?

Together, these ablations evaluate whether **pRisk-Pressure** captures meaningful structural differences in robustness beyond conventional success rates and whether probabilistic risk likelihood provide a more faithful characterization of real-world exploitability under bounded adversarial persistence.

11 LLM Use Declaration

Used ChatGPT for summarizing, rephrasing, brainstorming, and LaTeX formating.

References

Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, J Zico Kolter, Matt Fredrikson, Yarin Gal, and Xander Davies. Agentharm: A benchmark for measuring harmfulness of LLM agents. In *The Thirteenth International Conference on Learning Representations*, 2025. URL `https://openreview.net/forum?id=AC5n7xHuR1`.

Riccardo Cantini, Alessio Orsino, Massimo Ruggiero, and Domenico Talia. Benchmarking adversarial robustness to bias elicitation in large language models: Scalable automated assessment with llm-as-a-judge. *Machine Learning*, 114(11):249, 2025.

Nicholas Carlini, Milad Nasr, Christopher A Choquette-Choo, Matthew Jagielski, Irena Gao, Anas Awadalla, Pang Wei Koh, Daphne Ippolito, Katherine Lee, Florian Tramer, et al. Are aligned neural networks adversarially aligned?, 2024. *URL https://arxiv. org/abs/2306.15447*, 2023.

Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwag, Edgar Dobriban, Nicolas Flammarion, George J Pappas, Florian Tramer, et al. Jailbreakbench: An open robustness benchmark for jailbreaking large language models. *Advances in Neural Information Processing Systems*, 37:55005–55029, 2024.

Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. In *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, pages 23–42. IEEE, 2025.

Moussa Koulako Bala Doumbouya, Ananjan Nandi, Gabriel Poesia, Davide Ghilardi, Anna Goldie, Federico Bianchi, Dan Jurafsky, and Christopher D Manning. h4rm3l: A language for composable jailbreak attack synthesis. In *The Thirteenth International Conference on Learning Representations*, 2025. URL `https://openreview.net/forum?id=zZ8fgXHkXi`.

Isack Lee and Haebin Seong. Biasjailbreak:analyzing ethical biases and jailbreak vulnerabilities in large language models, 2025. URL `https://arxiv.org/abs/2410.13334`.

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, et al. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*, 2024.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does llm safety training fail? *Advances in neural information processing systems*, 36:80079–80110, 2023.

Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel, Yilun Zhao, Roy Bar-Haim, Arman Cohan, and Michal Shmueli-Scheuer. Survey on evaluation of llm-based agents. *arXiv preprint arXiv:2503.16416*, 2025.

Tongxin Yuan, Zhiwei He, Lingzhong Dong, Yiming Wang, Ruijie Zhao, Tian Xia, Lizhen Xu, Binglin Zhou, Fangqi Li, Zhuosheng Zhang, et al. R-judge: Benchmarking safety risk awareness for llm agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1467–1490, 2024.

Zhexin Zhang, Shiyao Cui, Yida Lu, Jingzhuo Zhou, Junxiao Yang, Hongning Wang, and Minlie Huang. Agent-safetybench: Evaluating the safety of llm agents, 2025. URL `https://arxiv.org/abs/2412.14470`.

Weikang Zhou, Xiao Wang, Limao Xiong, Han Xia, Yingshuang Gu, Mingxu Chai, Fukang Zhu, Caishuang Huang, Shihan Dou, Zhiheng Xi, et al. Easyjailbreak: A unified framework for jailbreaking large language models. *arXiv preprint arXiv:2403.12171*, 2024.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.